IS6400 Business Data Analytics

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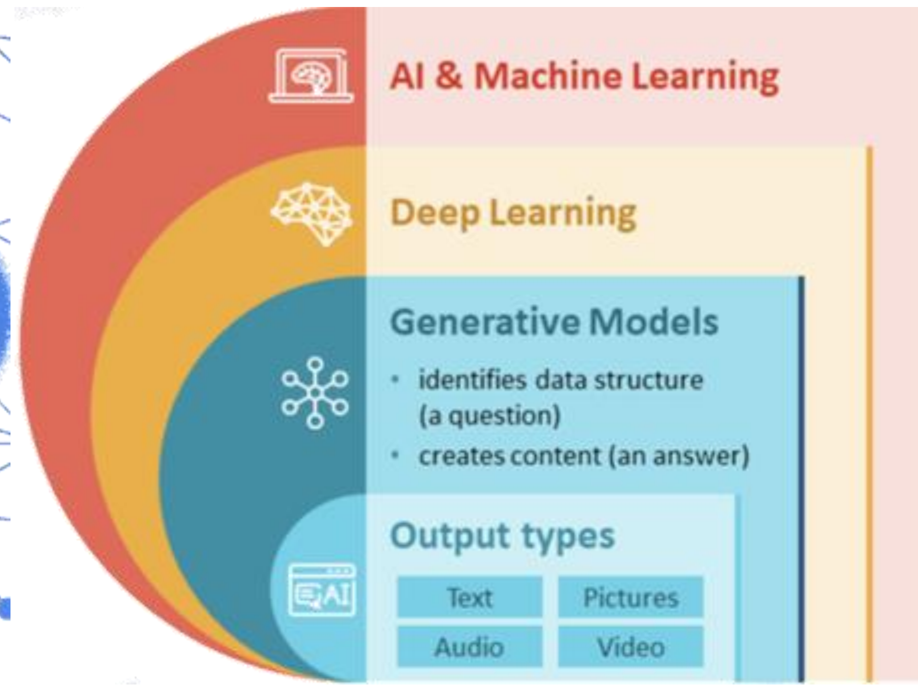
City University of Hong Kong

Lecture 7 Introduction to Generative AI

- **Overview**
- **Generative Models**
- Prompt Strategies
- Application & Challenges

Overview

❖ What is Generative AI ?



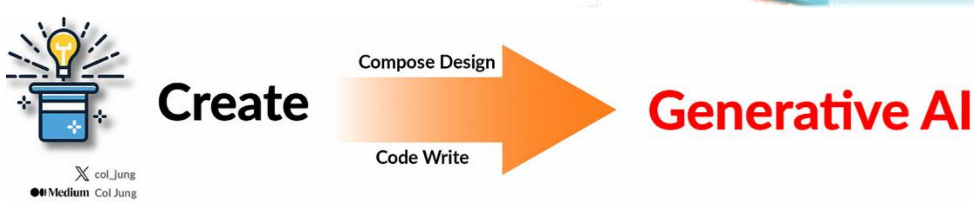


Image source: How Big Companies are Scrambling to Adopt Generative AI

Overview

❖ What is Generative AI ?

- Generative AI models use neural networks to identify the patterns and structures within existing data to generate new and original content.

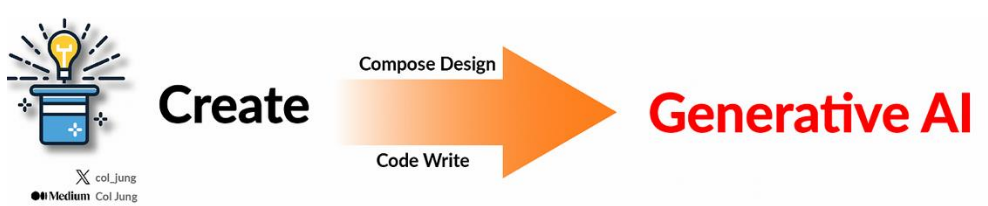


Image source: How Big Companies are Scrambling to Adopt Generative AI

Overview

❖ Generative AI

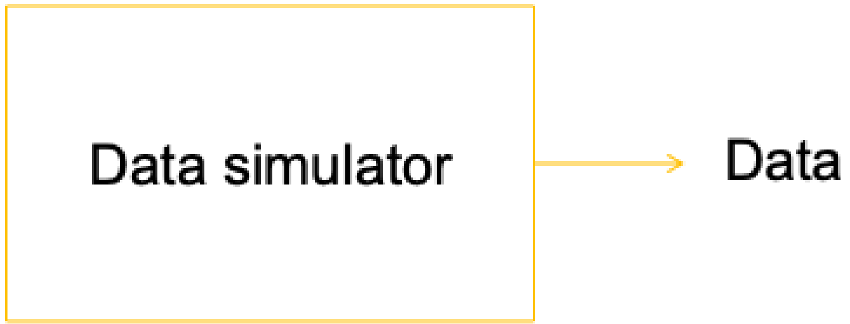
Image x  → A probability distribution p(x) → scalar probability p(x)

It is generative because **sampling from p(x) generates new images**

 . . . 

Overview

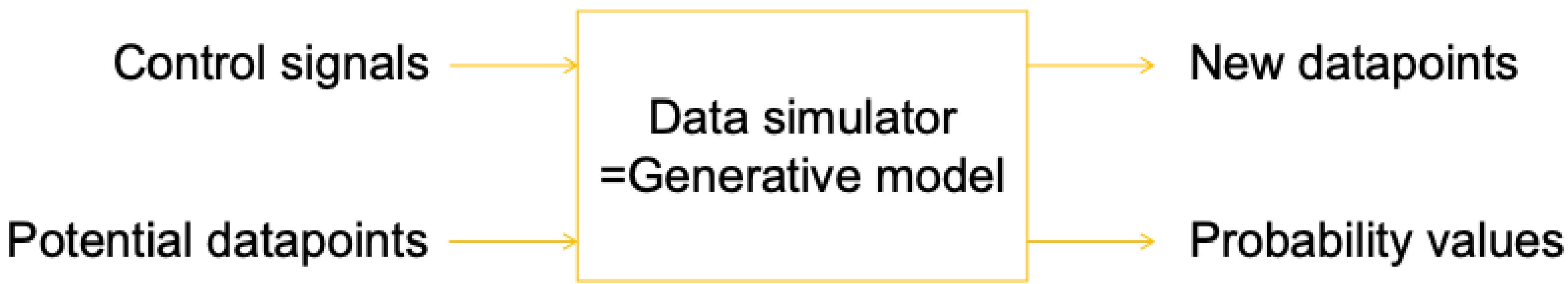
❖ **Understand Generative AI as Data Simulator**



Reference source: Stanford CS236 DeepGenerativemodels

Overview

❖ Understand Generative AI as Data Simulator



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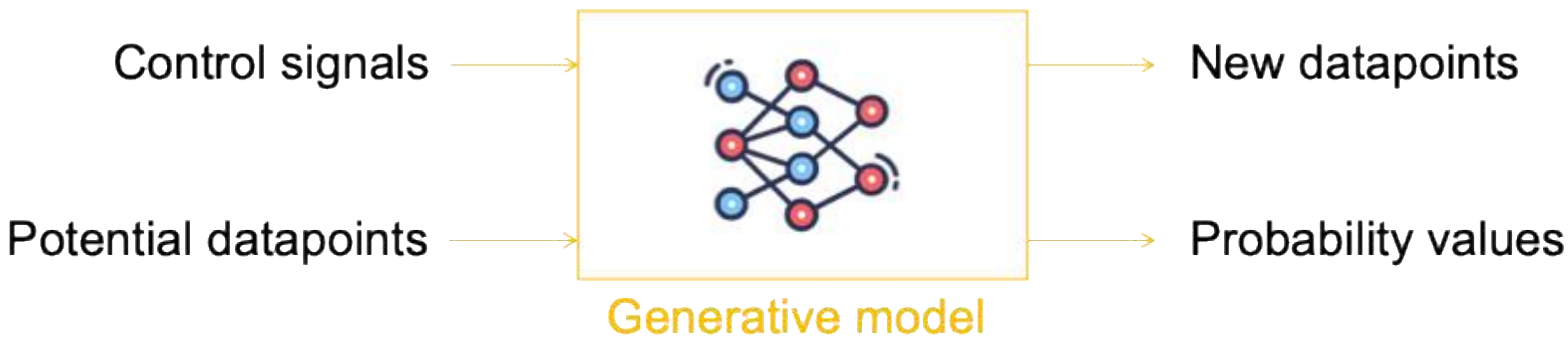
Overview

❖ Understand Generative AI as Data Simulator

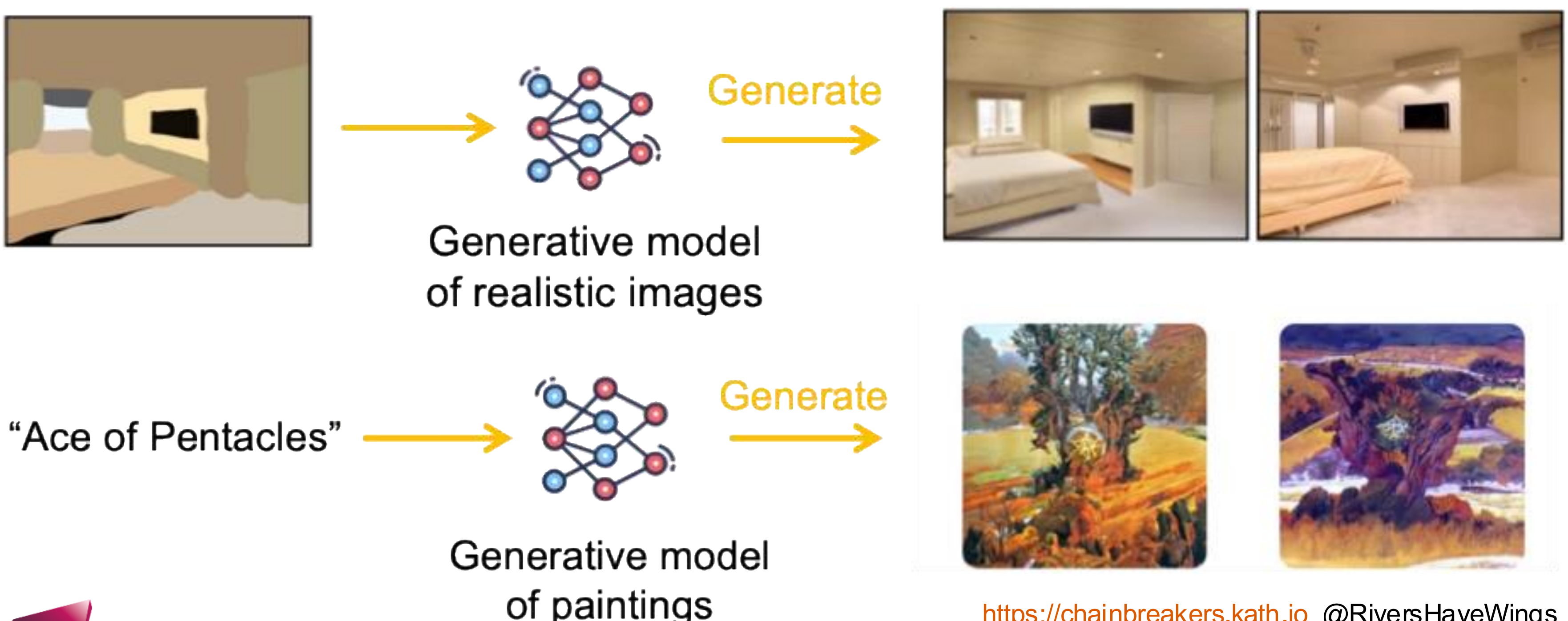


Reference source: Stanford CS236 DeepGenerativemodels

Overview

❖ Understand Generative AI as Data Simulator

- Examples



https://chainbreakers.kath.io @RiversHaveWings

Reference source: Stanford CS236 DeepGenerativemodels

Overview

❖ Understand Generative AI as Data Simulator

- Examples: text2image

DALL·E >

An astronaut riding a horse

Here you go! Two takes on an astronaut riding a horse across a surreal alien landscape. Let me know if you want to tweak the vibe—more dramatic lighting, different planet colors, or even a space cowboy theme! 🌌🐎🚀

Overview

❖ Understand Generative AI as Data Simulator

- Examples: text2image

泰迪熊穿着戏服，站在太和殿前唱京剧

A teddy bear, wearing a costume, is standing in front of the Hall of Supreme Harmony and singing Beijing opera

Overview

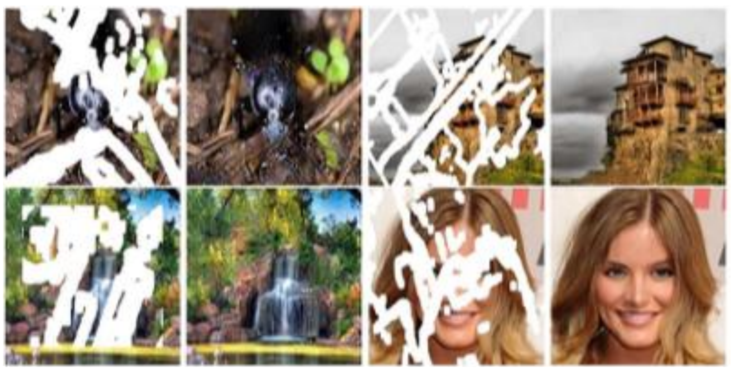
❖ Understand Generative AI as Data Simulator

- Examples: Photo Inverse

P(high resolution | low resolution)

Menon et al, 2020

P(full image| mask)



Liu al, 2018

P(color image| greyscale)

Antic, 2020

Reference source: Stanford CS236 DeepGenerativemodels

Overview

❖ Understand Generative AI as Data Simulator

- Examples: Language Generation

To get an A+ in Artificial Intelligence for Business Applications, students have to

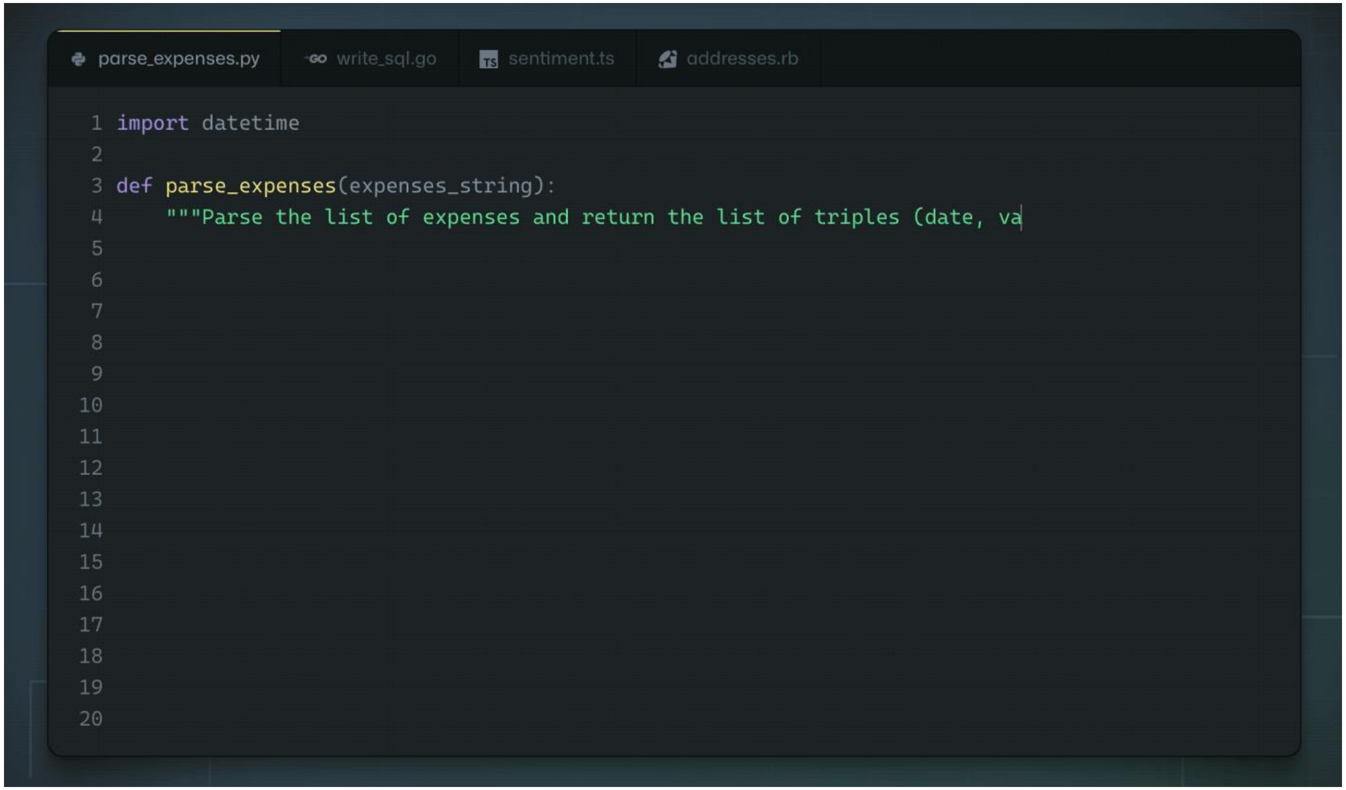
To get an A+ in **Artificial Intelligence for Business Applications**, students typically have to:

1. **Demonstrate Deep Understanding**
   - Master both technical AI concepts (e.g., machine learning models, neural networks, NLP) and their business applications (e.g., customer segmentation, demand forecasting, personalization).
   - Clearly explain the rationale behind choosing specific AI tools for given business problems.
2. **Apply AI to Real-World Business Problems**
   - Successfully complete case studies and projects that involve identifying a business need, selecting the right AI approach, implementing a prototype or analysis, and providing actionable insights.
   - Use clean, well-structured code and interpretable visualizations.
3. **Showcase Critical Thinking and Innovation**
   - Go beyond standard methods—propose creative solutions or enhancements.
   - Consider ethical, interpretability, and scalability issues in AI deployment.
4. **Perform Well in Assignments, Exams, and Participation**
   - Score highly on quizzes, midterms, and final exams that test both conceptual and applied knowledge.
   - Engage actively in class discussions, bring business-relevant examples, and participate in group activities or debates.
5. **Deliver an Outstanding Final Project**
   - Choose a meaningful, well-scoped business problem.
   - Justify the choice of data, algorithms, and evaluation metrics.
   - Demonstrate end-to-end pipeline thinking: from data understanding to decision-making.
   - Deliver a polished report and/or presentation, possibly with a dashboard or working prototype.
6. **Communicate Clearly and Professionally**

Overview

❖ Understand Generative AI as Data Simulator

- Examples: Code Generation



Overview

❖ Traditional AI Vs. Generative AI

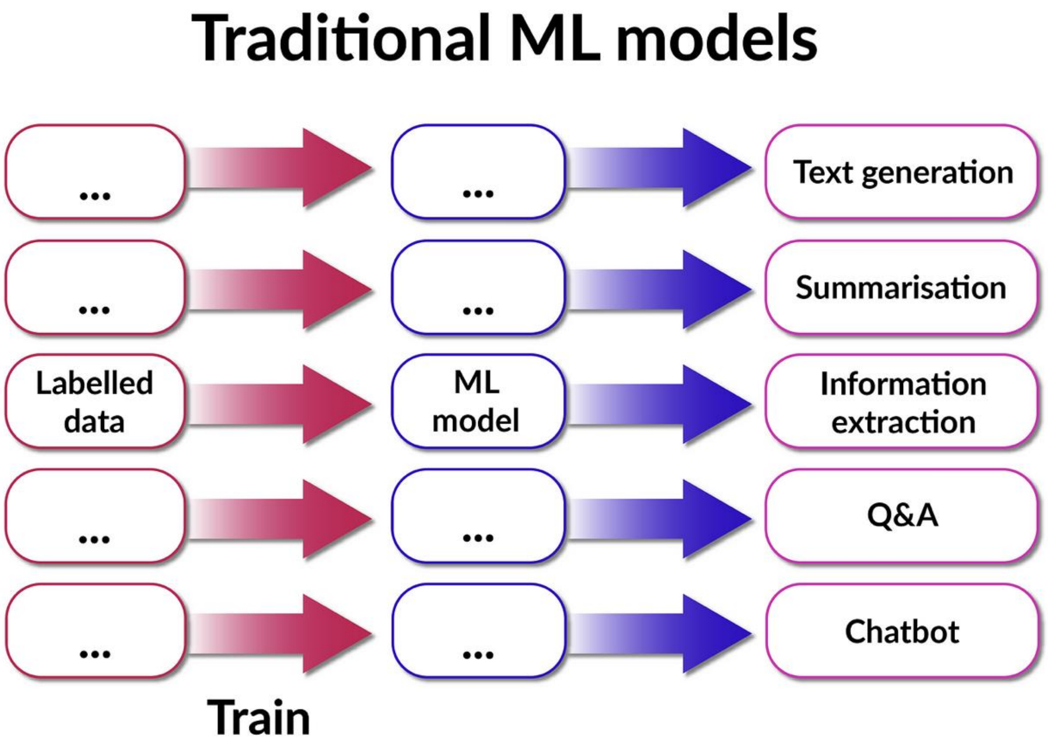
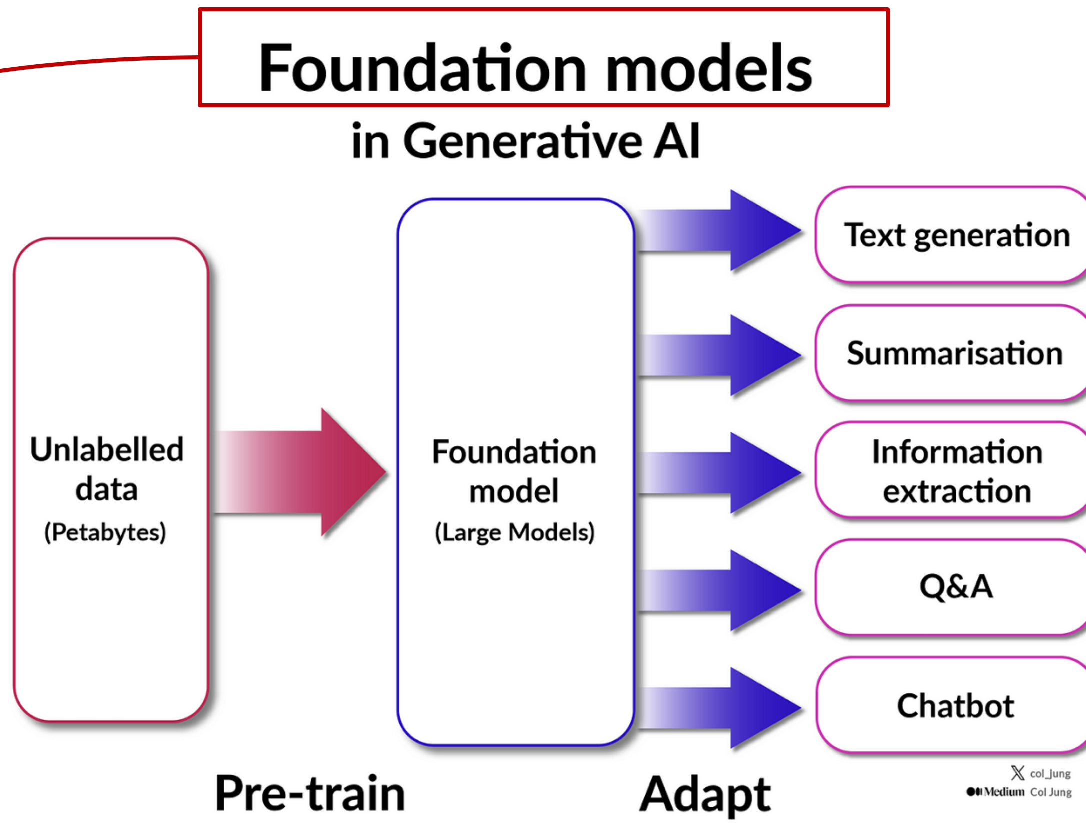


Image source: How Big Companies are Scrambling to Adopt Generative AI

Overview

❖ Traditional AI Vs. Generative AI

- GAN(2014).
- Diffusion(2015)
- Transformer(2017)
  - GPTs(2018-2022)
  - DeepSeek(2025)



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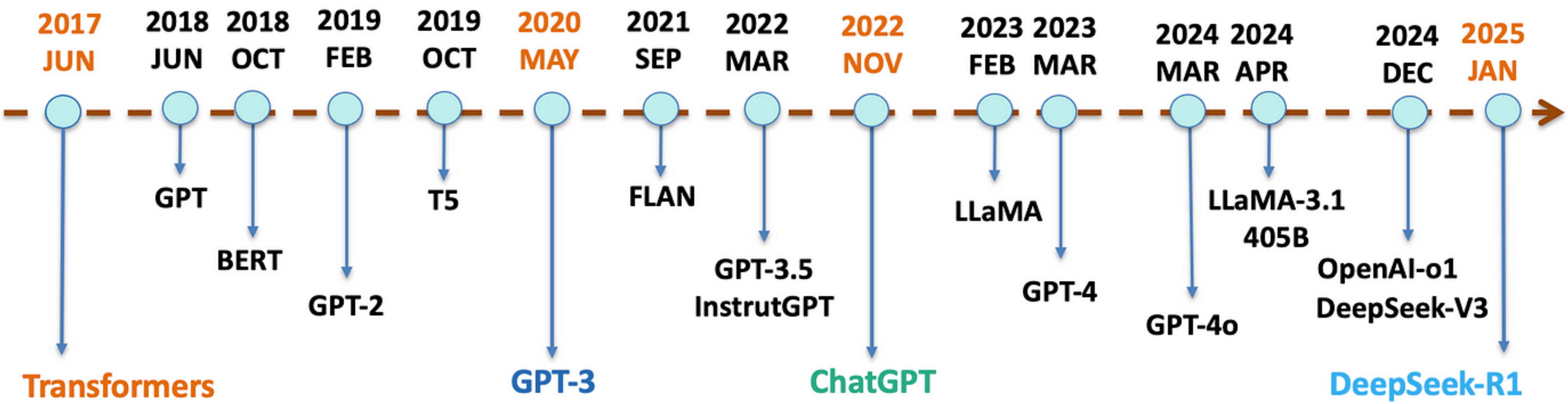
Lecture 11 Introduction to Generative Model

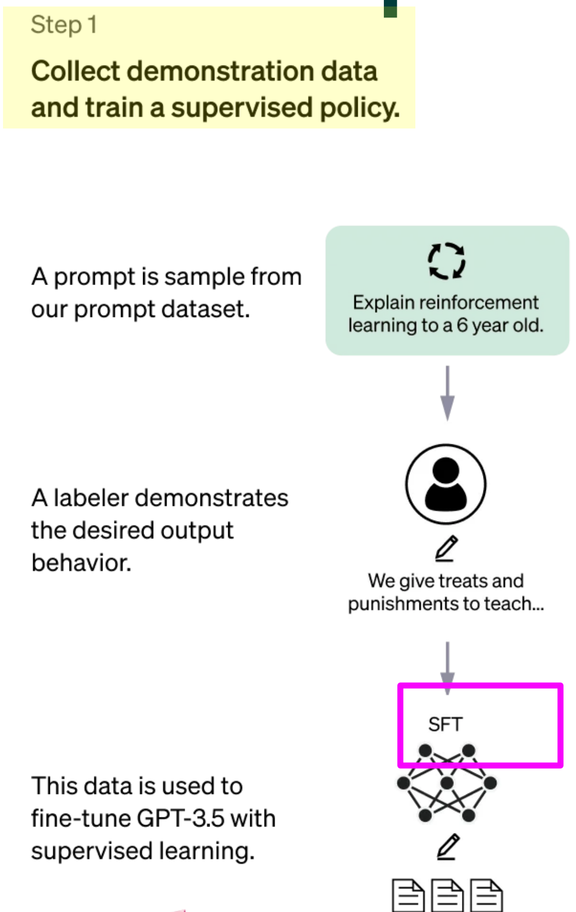
- Overview
- **Generative Models**
- Prompt Strategies
- Application & Challenges

A Brief History of LLMs



Generative AI Products

❖ Examples: ChatGPT



- Trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant.

Generative AI Products

❖ Examples: ChatGPT

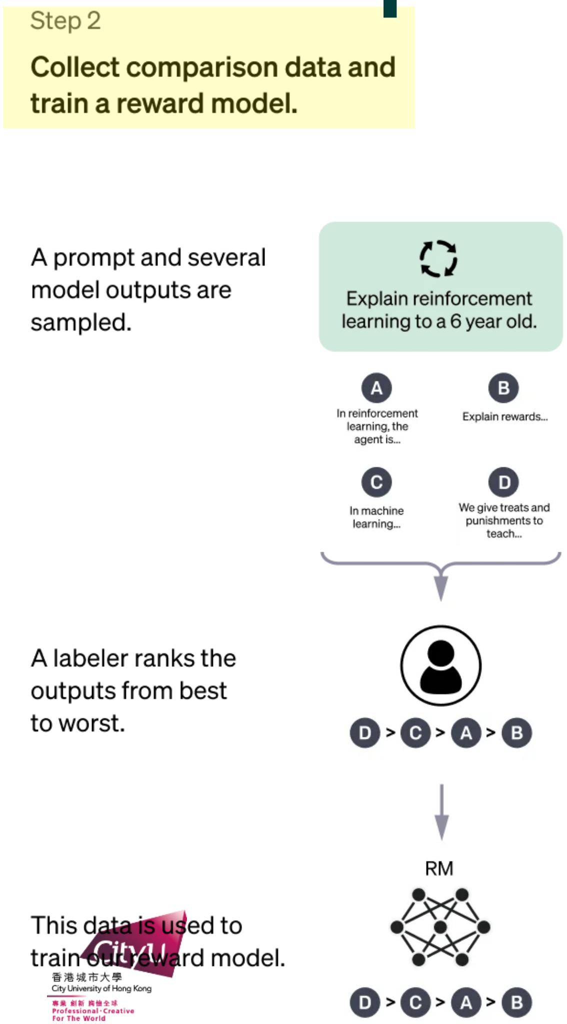
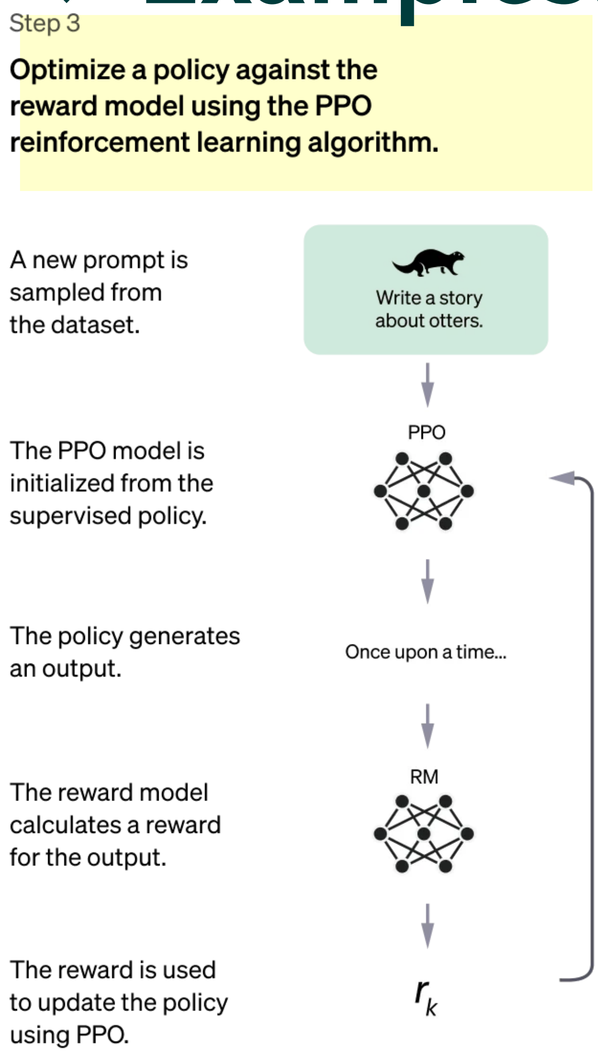


Image source: Introducing ChatGPT

- **comparison data:** consists of two or more model **responses** ranked by quality.
  - Took conversations that AI trainers had with the chatbot.
  - Randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them

Generative AI Products

❖ Examples: ChatGPT



- Using these reward models
- fine-tune the model using Proximal Policy Optimization.
- performed several iterations of this process.

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Lecture 11: Introduction to Generative Model

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- Generative Models
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Prompt Strategies

❖ Prompt Engineering

- **Prompt engineering** is structuring or crafting an instruction to produce the best possible output from a generative artificial intelligence model.
- A prompt is natural language text describing an AI's task

Prompt Strategies

❖ Strategies for Prompting

1. Write clear instructions.

- Models can't read your mind. If outputs are too long, ask for brief replies.
- If outputs are too simple, ask for expert-level writing. If you dislike the format, demonstrate the format you'd like to see.
- The less the model has to guess what you want, the more likely you'll get it.

Worse	Better
How do I add numbers in Excel?	How do I add up a row of dollar amounts in Excel? I want to do this automatically for a whole sheet of rows with all the totals ending up on the right in a column called "Total".
Who's president?	Who was the president of Mexico in 2021, and how frequently are elections held?
Write code to calculate the Fibonacci sequence.	Write a TypeScript function to efficiently calculate the Fibonacci sequence. Comment the code liberally to explain what each piece does and why it's written that way.

Image source: Prompt engineering

Prompt Strategies

❖ Strategies for Prompting

2. Provide reference text.

- Language models can confidently invent fake answers, especially when asked about esoteric topics or for citations and URLs.
- In the same way that a sheet of notes can help a student do better on a test,
- providing reference text to these models can help in answering with fewer fabrications

Image source: Prompt engineering

Prompt Strategies

❖ Strategies for Prompting

3. Split complex tasks into simpler subtasks.

- Complex tasks tend to have higher error rates than simpler tasks.
- Furthermore, complex tasks can often be re-defined as a workflow of simpler tasks in which the outputs of earlier tasks are used to construct the inputs to later tasks.

Image source: Prompt engineering

Prompt Strategies

❖ Strategies for Prompting

4. Give models time to "think"

- You might not know it instantly if asked to multiply 17 by 28, but you can still work it out with time.
- Similarly, models make more reasoning errors when trying to answer right away rather than taking time to work out an answer.
- Asking for a "chain of thought" before an answer can help the model reason its way toward correct answers more reliably.

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Image source: Prompt engineering

Prompt Strategies

❖ Strategies for Prompting

5. Use external tools

- Compensate for the model's weaknesses by feeding it the outputs of other tools.
- For example,
  - a text retrieval system (sometimes called RAG or retrieval augmented generation) can tell the model about relevant documents.
  - A code execution engine like OpenAI's Code Interpreter can help the model do the math and run code.
- If a task can be done more reliably or efficiently by a tool rather than a language model, offload it to get the best of both.

Image source: Prompt engineering

Prompt Strategies

Strategies for Prompting

- "Give a man a fish"
  - Automates process of report generation and data visualization
    - Julius AI, Formula Bot, Mirror
  - Enhance data cleaning and integration
  - Support advanced predictive modeling, economics models
    - Oxford Economics – AI-assisted tools for macro/data analysis
- "Teach a man how to fish"
  - Instructor for Python learning, new software installation
  - Foster certain mindset, chain-of-thoughts

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Image source: Prompt engineering

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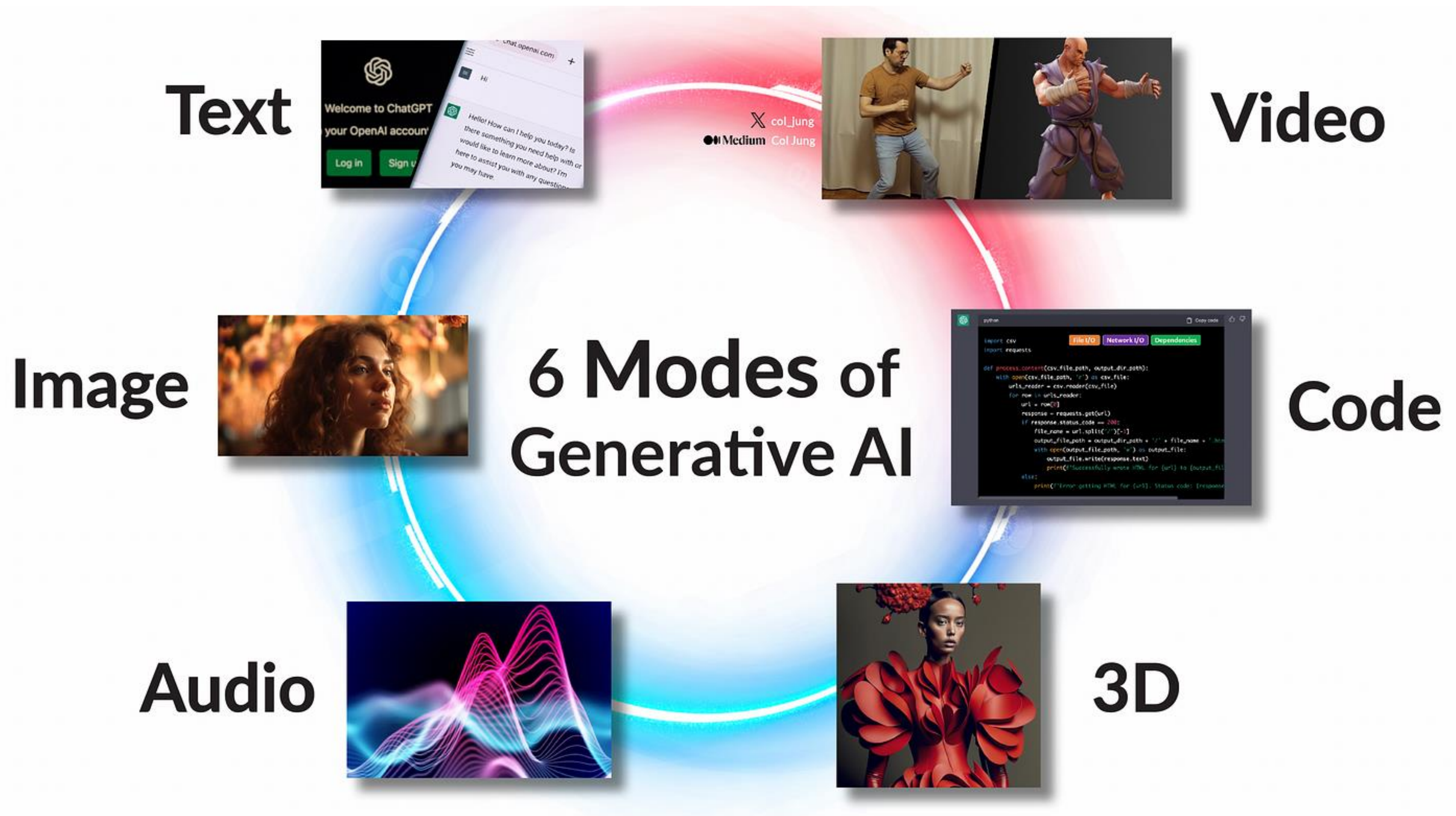
Applications in Industry



Challenges

❖ Hallucination

- Hallucinations are a critical problem for natural language generation systems using large language models (LLMs), such as ChatGPT or Gemini, because users cannot trust that any given output is correct.
- **Definition**
  - LLMs generate "**content that is nonsensical or unfaithful to the provided source content**," but they have come to include many failures of faithfulness and factuality.

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Challenges

❖ Hallucination

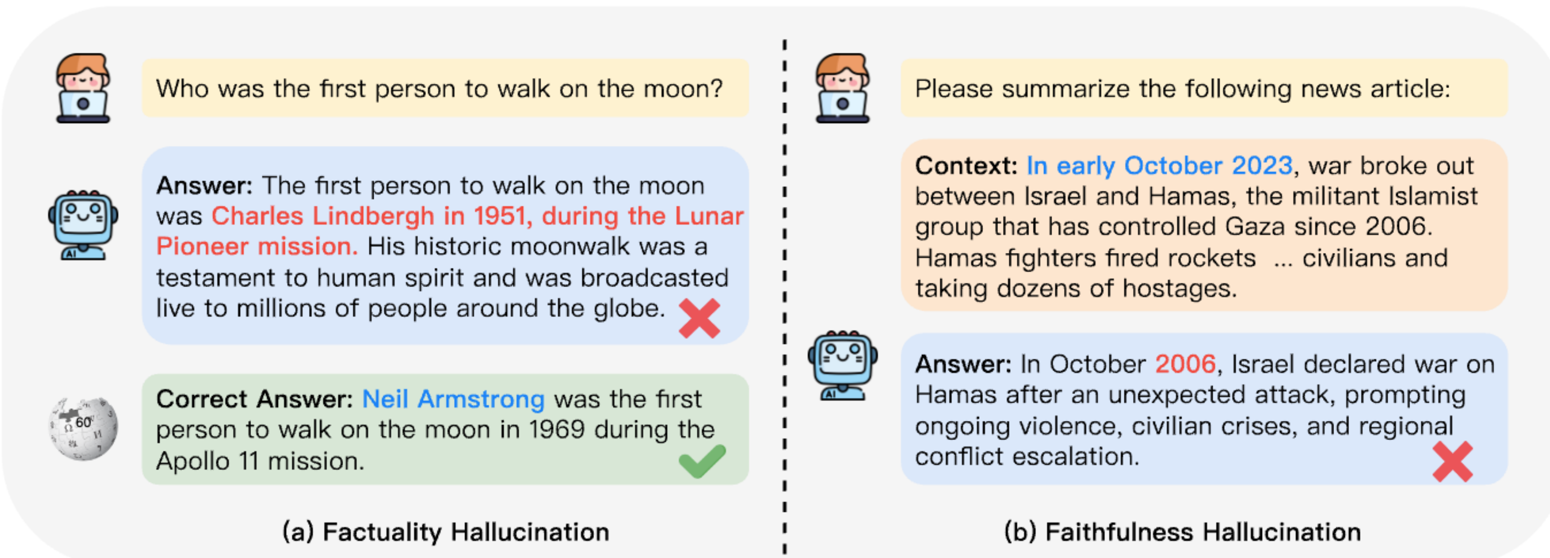

Figure 1: An intuitive example of LLM hallucination.

Image source: A Survey on Hallucination in Large Language Models

Challenges

❖ Model Randomness

- While randomness fuels creativity, it can also make the AI's outputs less predictable and consistent.
- This can be challenging when precise, repeatable outcomes are desired. For instance,
    - Unpredictable outputs could seriously affect sensitive applications like healthcare or financial services.

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Reference source: Generative AI Meets Responsible AI: Practical Challenges and Opportunities

Challenges

❖ Model robustness

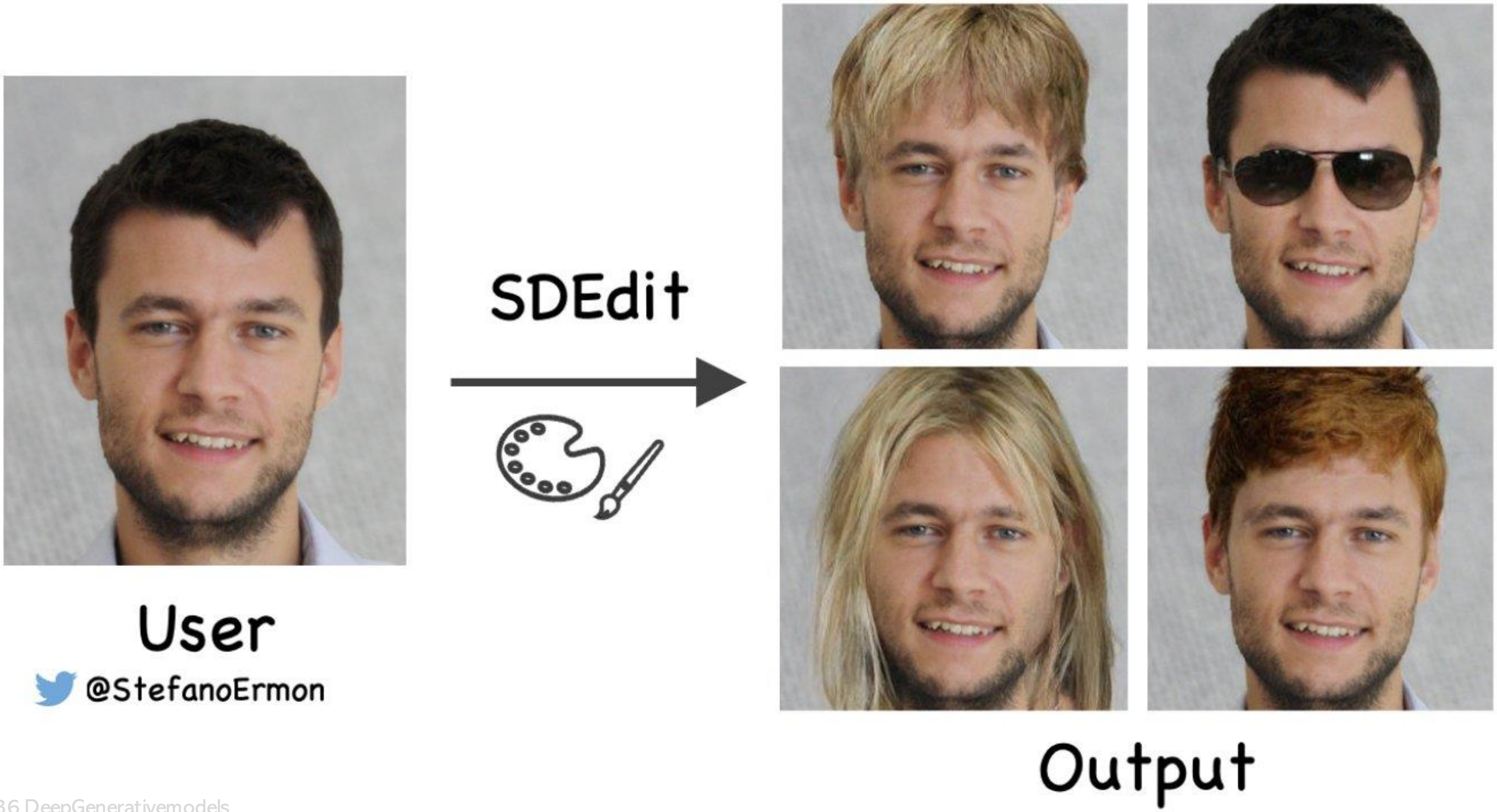
- LLMs often lack the ability to provide uncertainty estimates.
- Without knowledge of the extent of the model's confidence (or uncertainty), it becomes difficult for users to decide when the model's output can be trusted.

Reference source: Generative AI Meets Responsible AI: Practical Challenges and Opportunities

Challenges

❖ Cybercrime

Deepfake： Which image is real?



Reference source: Stanford CS236 DeepGenerativemodels

Challenges

❖ Bias and discrimination

- Generative AI models are often trained on large corpora of data, making it difficult to audit the training data for different types of biases, such as gender stereotypes.
- AI models trained on automatically collected web-scraped data have been shown to learn biases of sexual objectification, which can propagate to downstream applications.

Reference source: Generative AI Meets Responsible AI: Practical Challenges and Opportunities